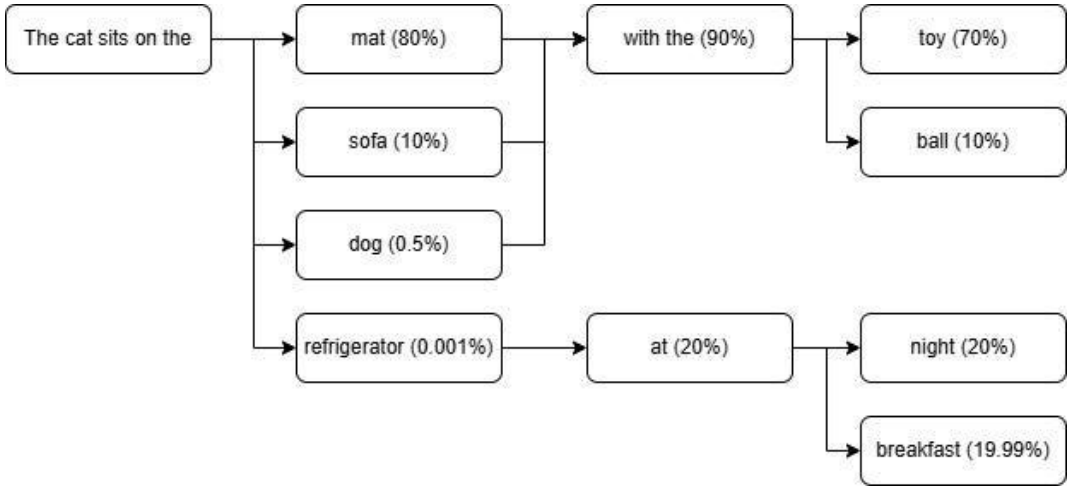
Лекция 12. Языковые модели. Рекуррентные нейронные сети. Применение к биологическим последовательностям

Фролов Дмитрий

Сириус, 2026

Языковая модель

Языковая модель — это модель, которая оценивает вероятность всей последовательности токенов целиком или ***вероятность следующего токена*** по предыдущему контексту.

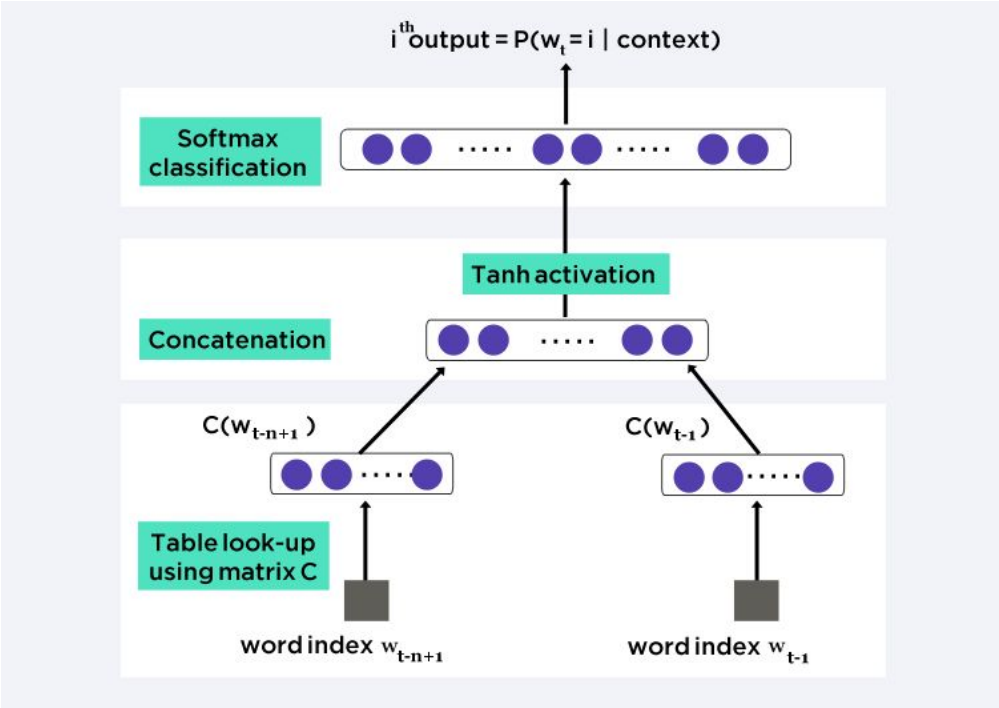


N-Gram (Марковская цепь)

$$P(y_1, y_2, \ldots, y_n) = P(y_1) \cdot P(y_2|y_1) \cdot P(y_3|y_1, y_2) \cdot \cdots \cdot P(y_n|y_1, \ldots, y_{n-1}) = \prod_{t=1}^{n} P(y_t|y_{<t})$$

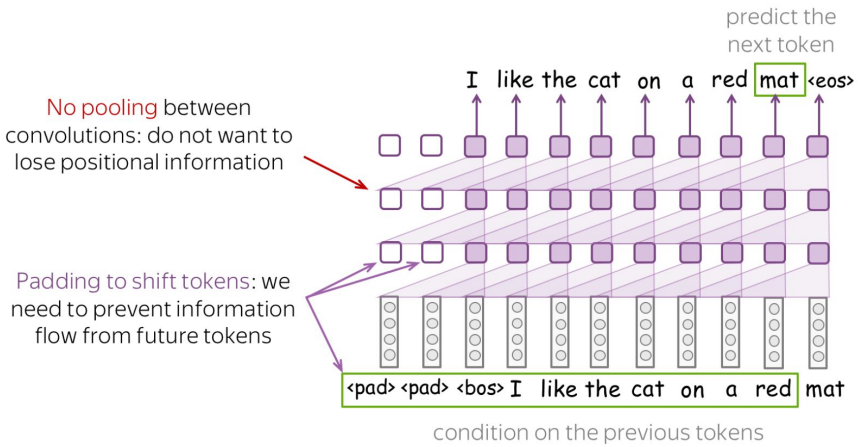
$$\begin{aligned}\widehat{\mathrm{P}}(\text{mat} \mid \text{cat on a}) \approx\ & \lambda_3 \mathrm{P}(\text{mat} \mid \text{cat on a}) + \\ & \lambda_2 \mathrm{P}(\text{mat} \mid \text{on a}) + \\ & \lambda_1 \mathrm{P}(\text{mat} \mid \text{a}) + \\ & \lambda_0 \mathrm{P}(\text{mat})\end{aligned}$$

Первая нейронная языковая модель

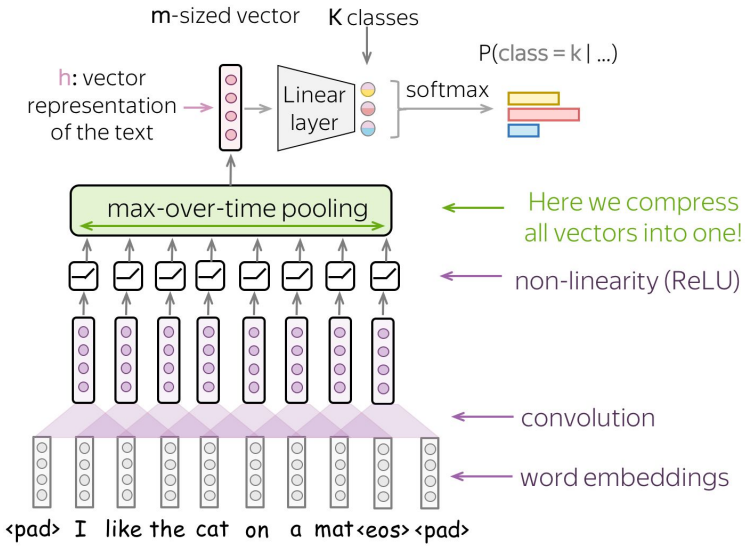


Bengio Y et al. A neural probabilistic language model. Journal of machine learning research. 2003;3(Feb):1137-55.

Сверточные сети для текста



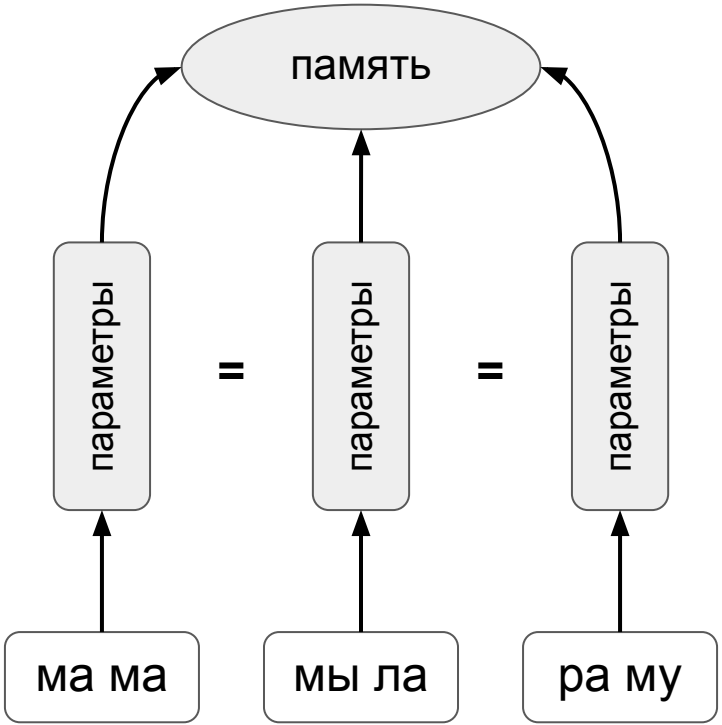
Временная/казуальная свертка



Классическая свертка

Images Source: https://lena-voita.github.io/nlp_course/models/convolutional.html

Нейросети нужна память



RNN обрабатывает последовательность итеративно

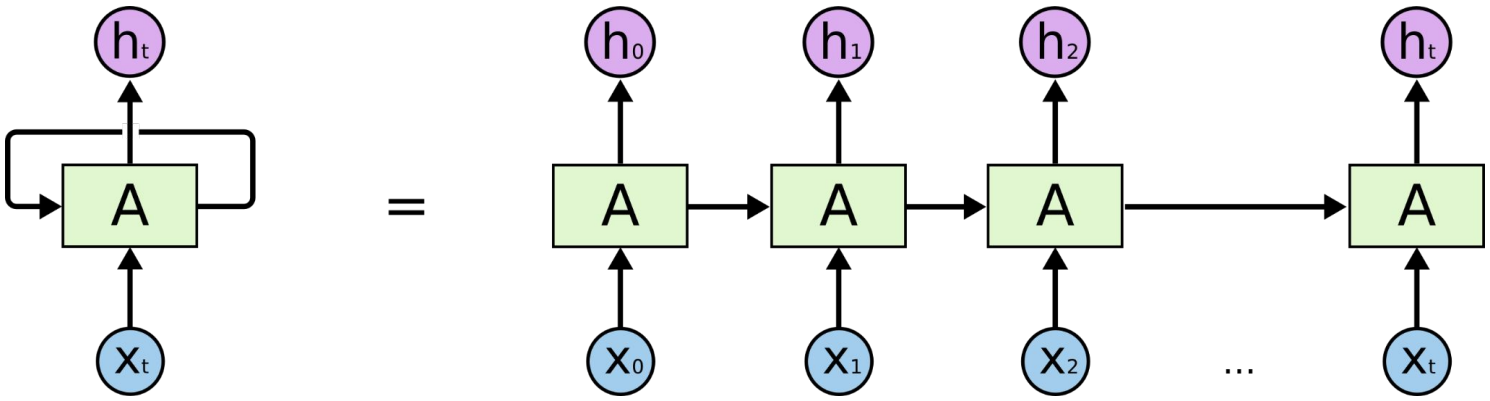
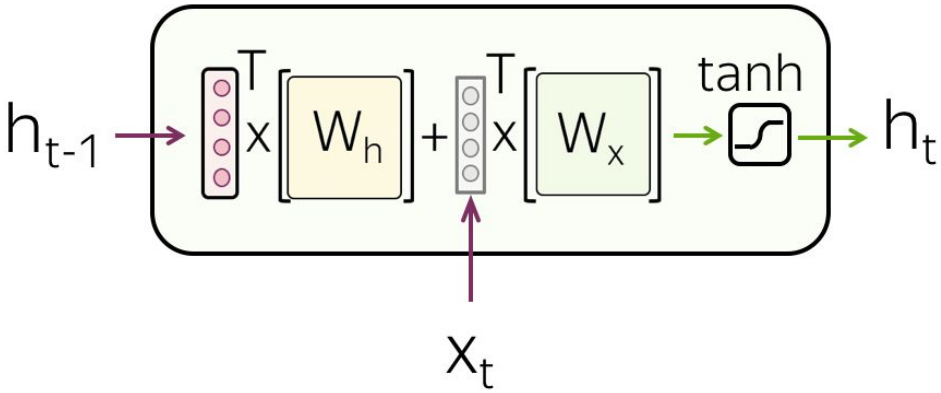


Image Source: https://colah.github.io/posts/2015-08-Understanding-LSTMs

Vanilla RNN (Сеть Элмана)

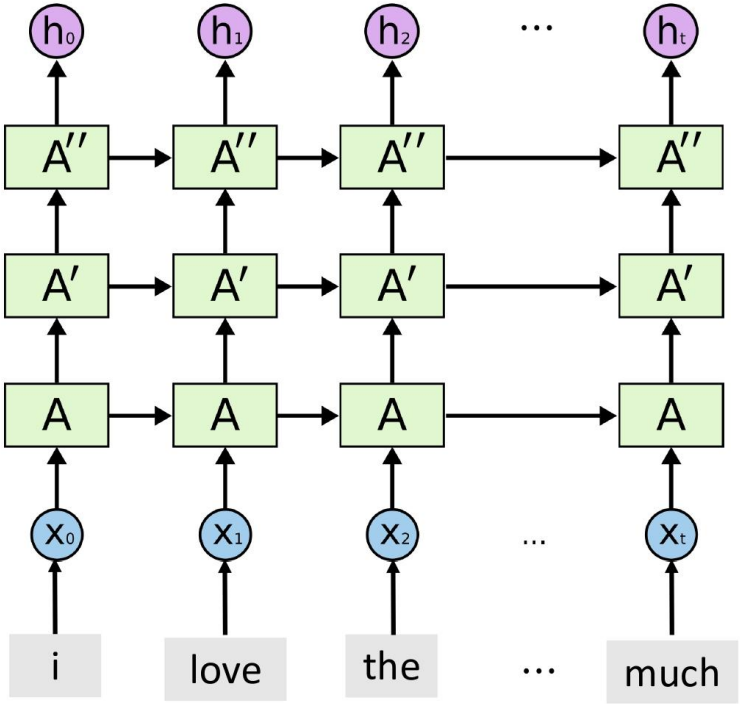


$$h_t = \tanh\left(h_{t-1}^\top W_h + x_t^\top W_x + b\right)$$

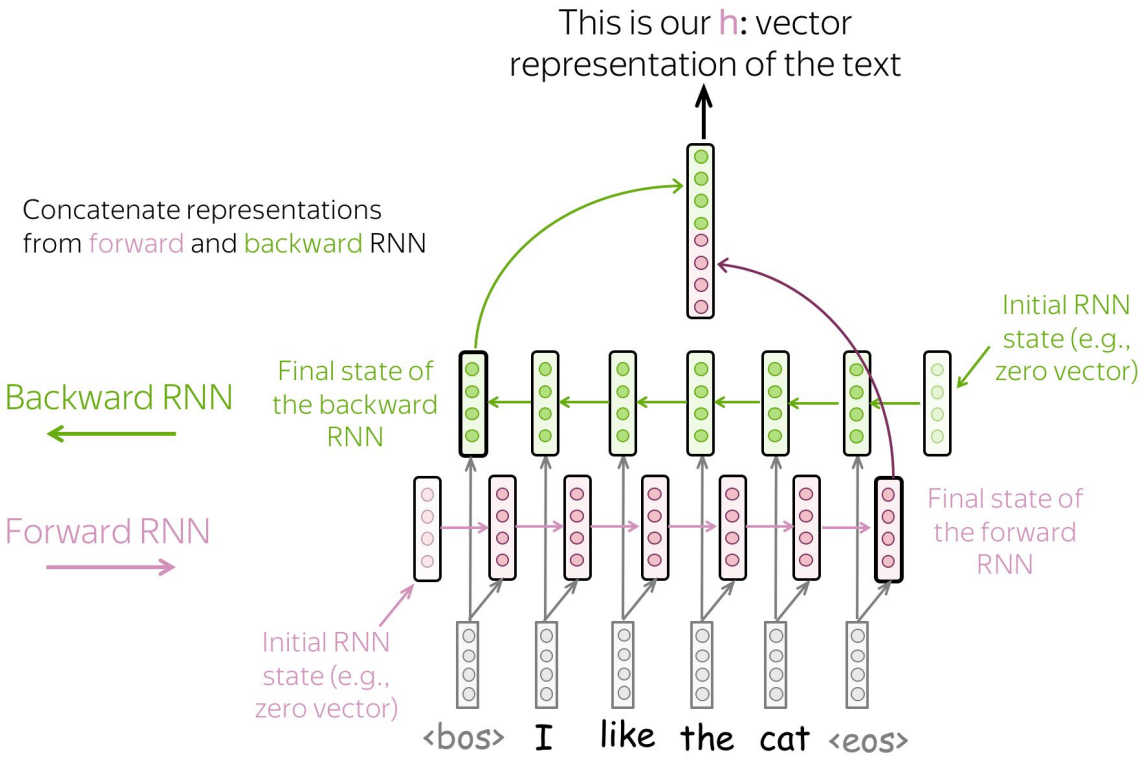
Elman JL. Finding structure in time. Cognitive science. 1990 Mar;14(2):179-211.

Werbos PJ. Backpropagation through time: what it does and how to do it. Proceedings of the IEEE. 1990;78(10):1550-60.

Stacked RNN



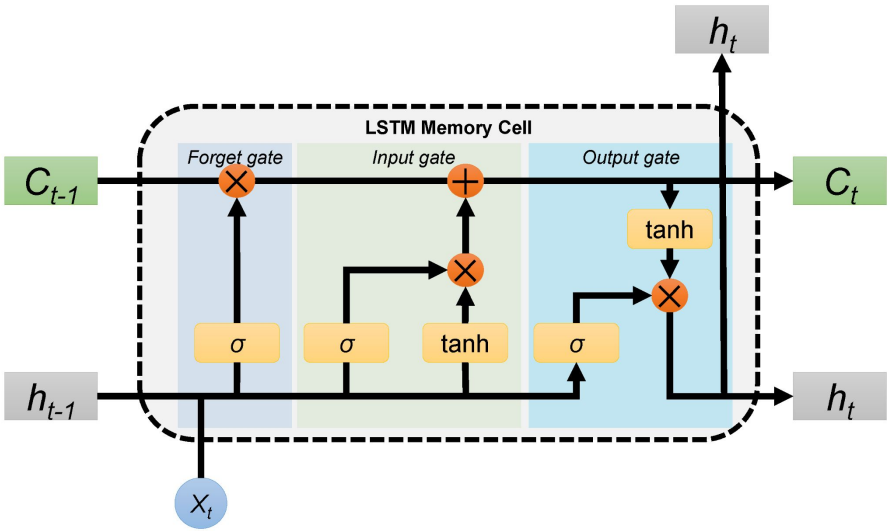
Bidirectional RNN



Schuster M, Paliwal KK. Bidirectional recurrent neural networks. IEEE transactions on Signal Processing. 1997 Nov 30;45(11):2673-81.

Image Source: https://lena-voita.github.io/nlp_course/text_classification.html

Long Short-Term Memory



$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

Что забыть?

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

Что запомнить?

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Обновляем память

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(c_t)$$

Выдаем информацию

Hochreiter S, Schmidhuber J. Long short-term memory. Neural computation. 1997 Nov 15;9(8):1735-80.

Gated Recurrent Unit

$$z_t = \sigma(W_z[x_t, h_{t-1}] + b_z)$$

Какую долю старой информации сохранить?

$$r_t = \sigma(W_r[x_t, h_{t-1}] + b_r)$$

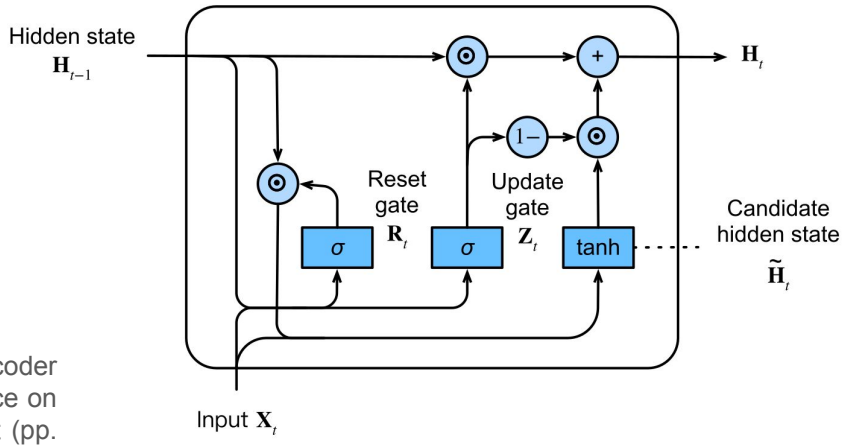
Какую долю старой информации использовать для обновления?

$$\tilde{h}_t = \tanh(W_h[x_t, r_t \odot h_{t-1}] + b_h)$$

Кандидат для обновления состояния

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t$$

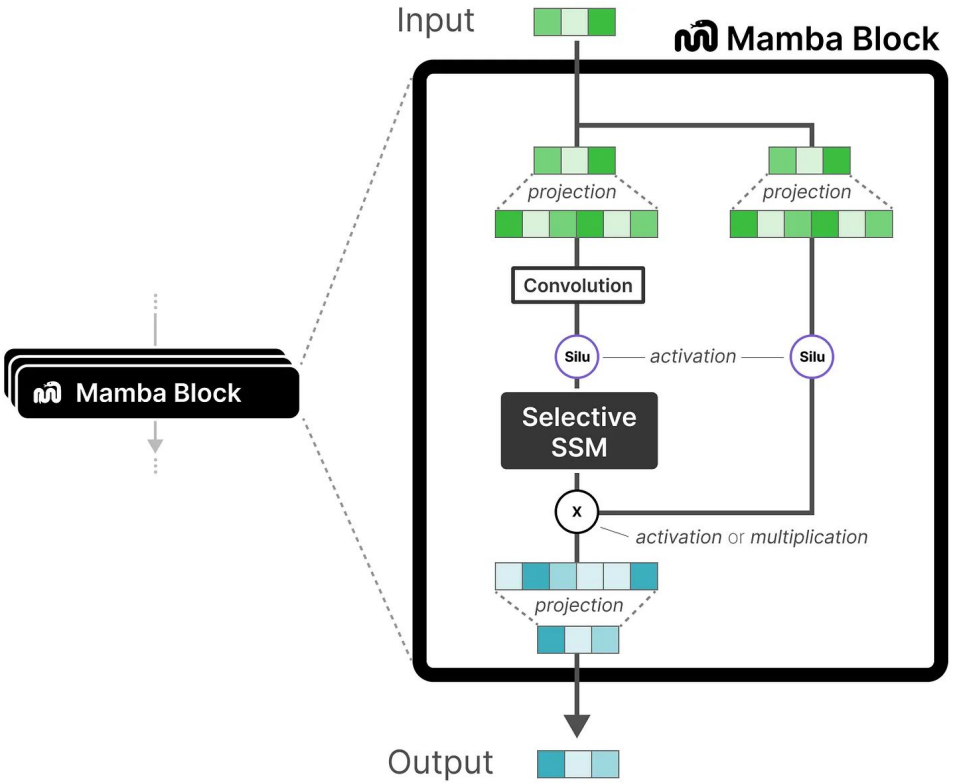
Обновляем скрытое состояние

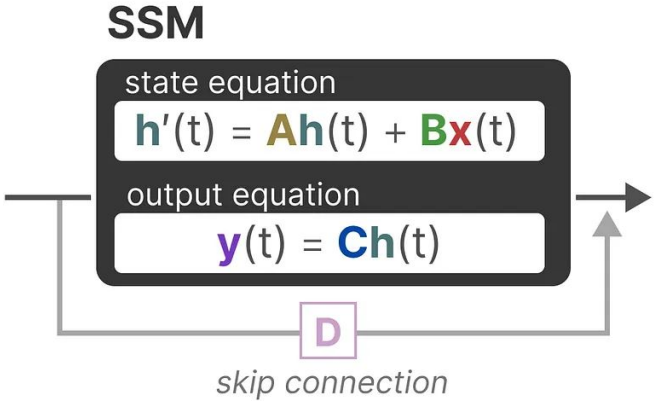


Cho K et al. Learning phrase representations using RNN encoder–decoder for statistical machine translation. InProceedings of the 2014 conference on empirical methods in natural language processing (EMNLP) 2014 Oct (pp. 1724-1734).

Image Source: https://d2l.ai/chapter_recurrent-modern/gru.html

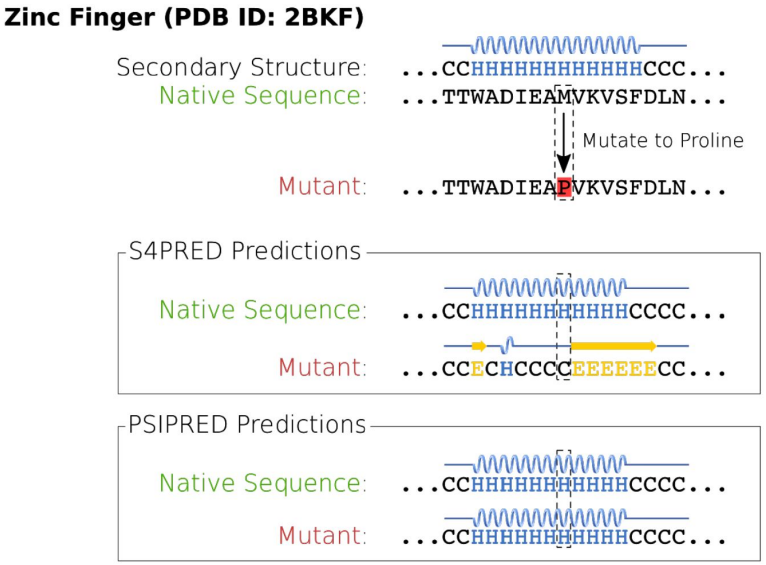
State Space Models и Mamba

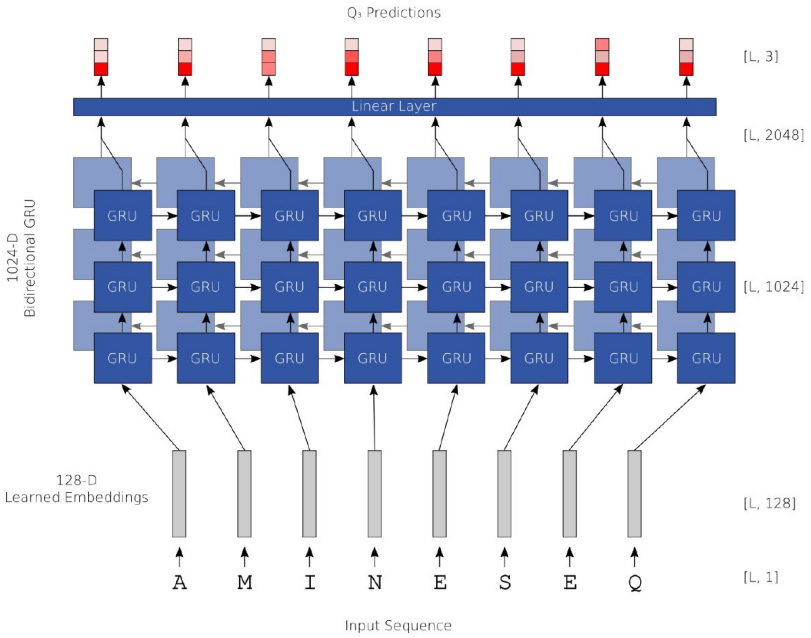




Gu A, Dao T. Mamba: Linear-time sequence modeling with selective state spaces. InFirst conference on language modeling 2024 May.
Images Source: https://newsletter.maartengrootendorst.com/p/a-visual-guide-to-mamba-and-state

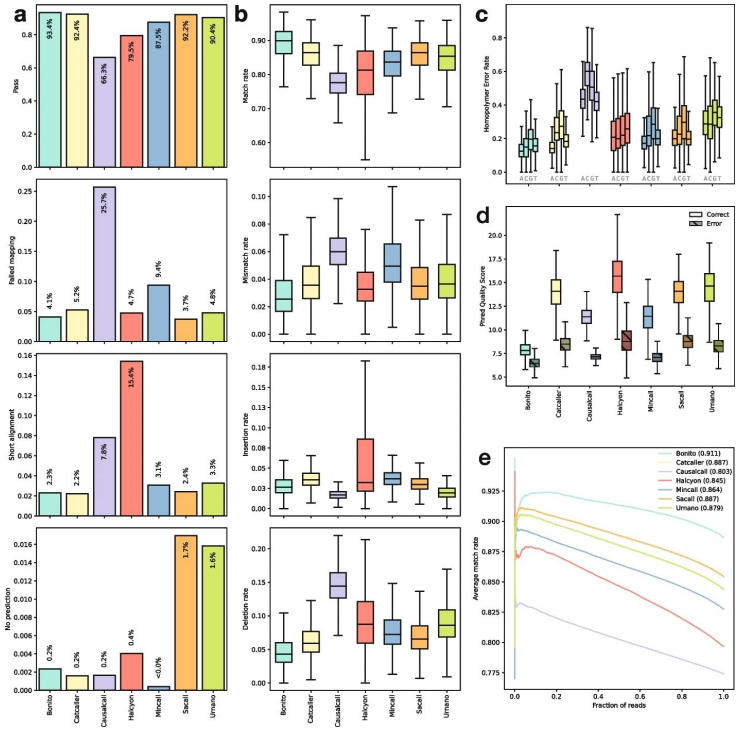
S4PRED — предсказание вторичной структуры белка

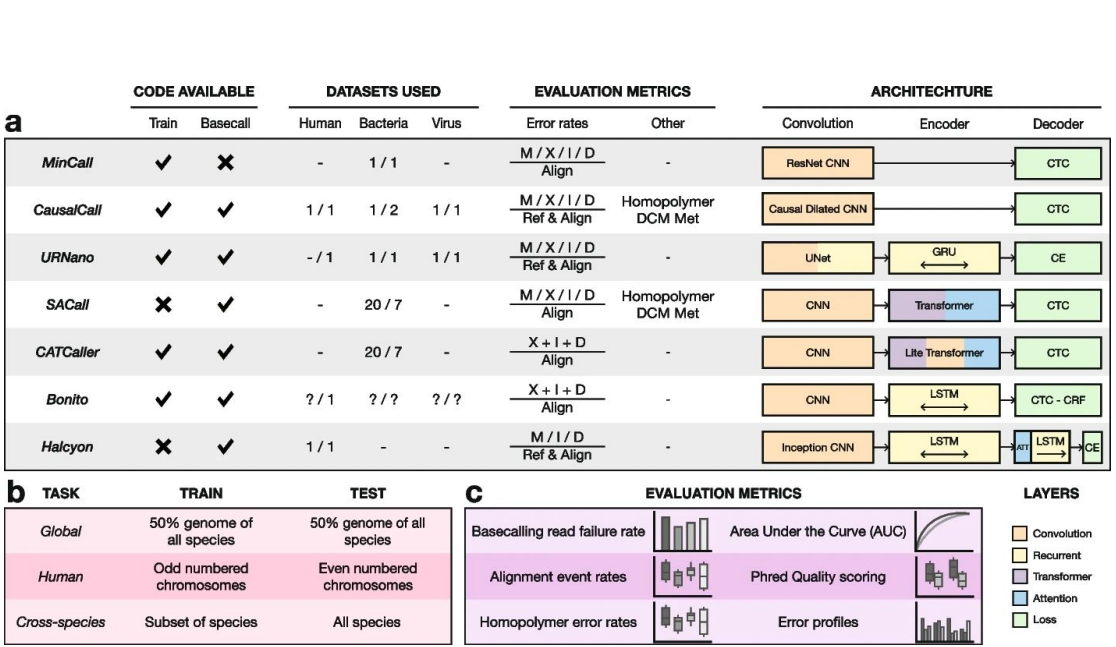




Moffat L, Jones DT. Increasing the accuracy of single sequence prediction methods using a deep semi-supervised learning framework. Bioinformatics. 2021 Nov 1;37(21):3744-51.

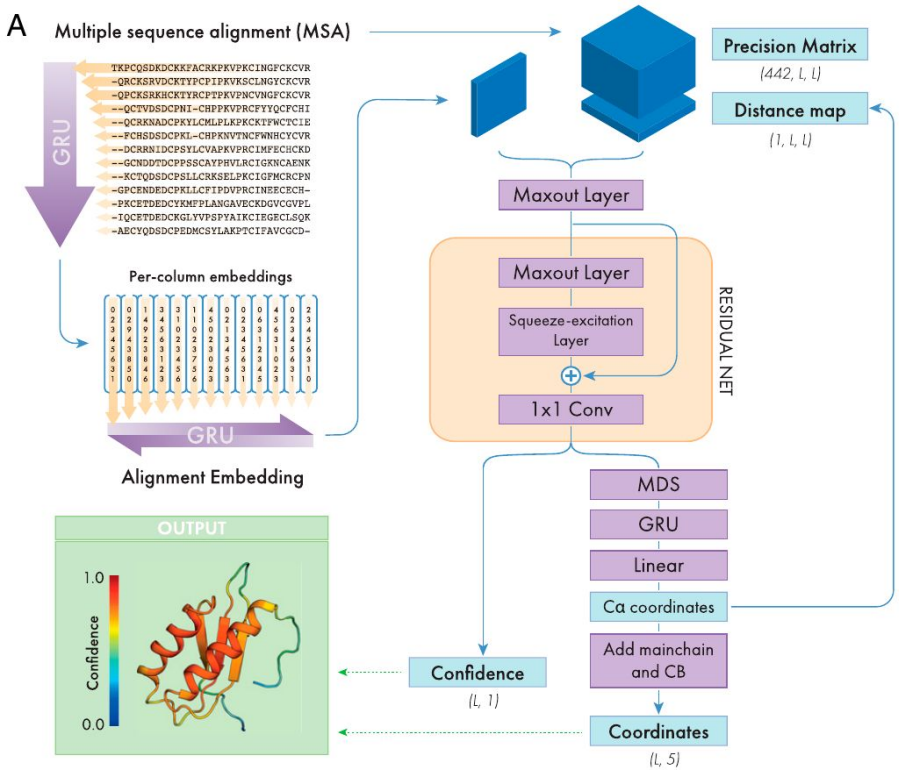
Бейзколлинг для нанопорового секвенирования

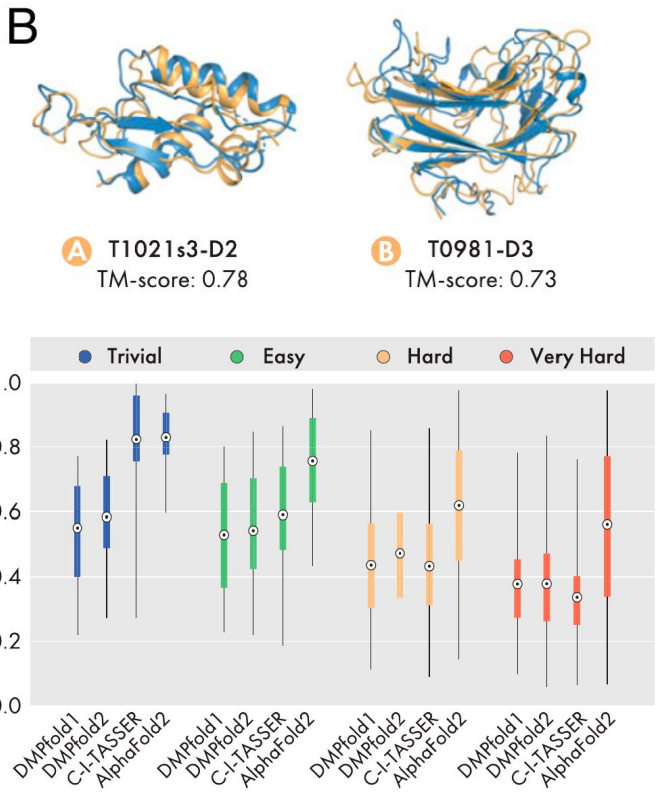




Pagès-Gallego M, De Ridder J. Comprehensive benchmark and architectural analysis of deep learning models for nanopore sequencing basecalling. Genome Biology. 2023 Apr 11;24(1):71.

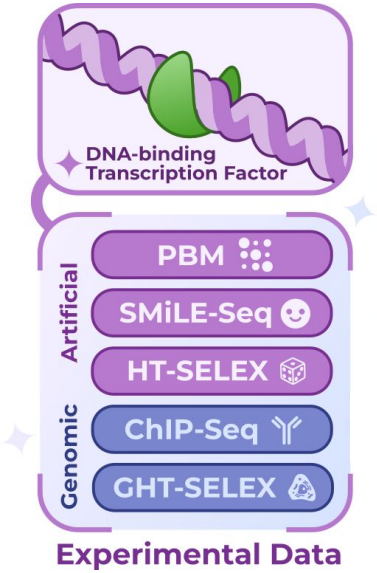
DMPFold2 — предсказание трехмерной структуры белка



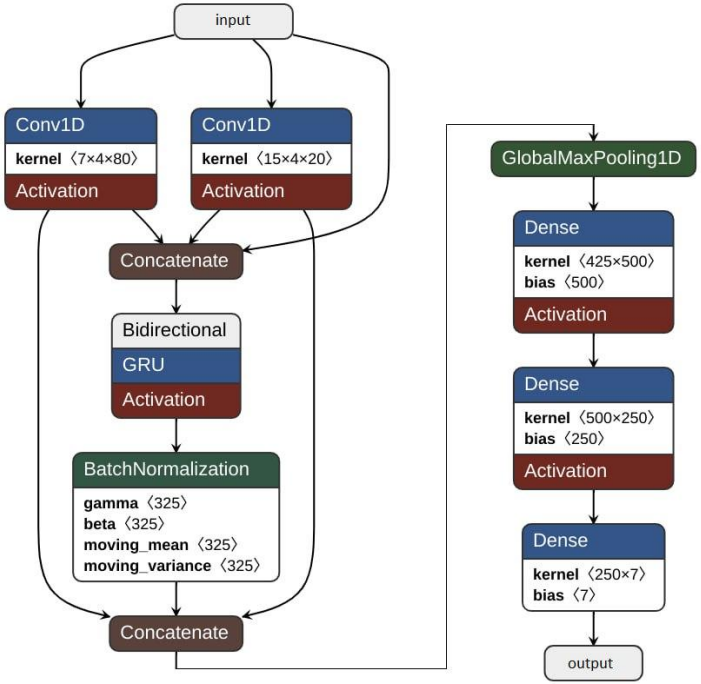


Kandathil SM et al. Ultrafast end-to-end protein structure prediction enables high-throughput exploration of uncharacterized proteins. Proceedings of the National Academy of Sciences. 2022 Jan 25;119(4):e2113348119.

Предсказание связывания TF на соревновании IBIS



	A2G ranks	
Name	AAA ↑	PWM
Biology Impostor	🏆	🏆
Salimov and Frolov Laboratory	🏆	8
mj	🏆	8
Natural Killer	4	8
callitmagic	5	🏆
Bench Pressers	6	🏆
sbi two	7	7
ChatGPTusers	8	4
The Motifvators	9	8



Gryzunov N et al. Inferring binding specificities of human transcription factors with the wisdom of crowds. bioRxiv. 2025 Nov 17.